

Credit Risk Prediction System

Milestone 4 Report: Model Development & Evaluation

Introduction

Milestone 4 focuses on building and evaluating machine learning models to predict loan default risk. The goal is to develop a high-performance model that can accurately classify borrowers as **safe** or **high-risk (default)**.

Objective

- To train multiple machine learning models
- To evaluate their performance using standard metrics
- To select the best-performing model
- To prepare the model for deployment

Dataset Description

- Dataset used: final_features.csv
- Target variable: default
 - 0 → Safe
 - 1 → Default Risk

Feature Categories

1. Numerical Features

- age
- loanamount
- creditscore
- monthsemployed

- numcreditlines
- interestrate
- loanterm
- dtiratio

2. Engineered Features

- loan_income_ratio
- credit_utilization
- interest_burden
- employment_stability
- high_dti_flag
- log_income
- income_loan_interaction

3. Encoded Categorical Features

- education (high school, master's, PhD)
- employment type (part-time, self-employed, unemployed)
- marital status
- loan purpose
- mortgage status
- cosigner

Methodology

1. Data Preprocessing

- Missing values handled
- One-hot encoding applied to categorical variables

- Data cleaned and formatted
- Outliers treated where required

2. Feature Engineering

New features were created to improve predictive power:

- Loan-to-income ratio → Financial burden
- Credit utilization → Credit usage behavior
- Interest burden → Loan repayment pressure
- Employment stability → Income reliability
- High DTI flag → Risk indicator
- Log transformation → Reduce skewness
- Interaction features → Capture relationships

3. Model Training

The following models were trained:

- Logistic Regression
- Random Forest Classifier
- Gradient Boosting Classifier

4. Model Evaluation

Each model was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC Score

5. Visualization

- Confusion Matrix
- ROC Curve
- Precision-Recall Curve
- Feature Importance Graph

These helped in understanding model performance and feature contribution.

6. Model Selection

- Best model selected based on:
 - High Recall (important for detecting defaulters)
 - High ROC-AUC score

7. Model Saving

The final model was saved as:

model.pkl

Outcome of Milestone 4

- A trained and optimized machine learning model
- Improved prediction accuracy and recall
- Model ready for real-world deployment
- Strong ability to identify high-risk borrowers

Conclusion

Milestone 4 successfully delivered a high-performance predictive model. The model captures complex financial relationships and provides reliable predictions for credit risk assessment.
